

Code generation gadget after prototype pollution

Code generation gadget after prototype pollution - dcodeIO, protobuf.js.

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Content

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protobuf.js: Code generation gadget after prototype pollution

- Advisory: GHSA-75px-5xx7-5xc7
- CVE: CVE-2026-44291
- Severity: high
- Published: 2026-05-12
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Affected

- `protobufjs` (npm): <= 7.5.5, fixed in 7.5.6
- `protobufjs` (npm): >= 8.0.0, <= 8.0.1, fixed in 8.0.2

Description

Summary

protobufjs used plain objects with inherited prototypes for internal type lookup tables used by generated encode and decode functions. If `Object.prototype` had already been polluted, those lookup tables could resolve attacker-controlled inherited properties as valid protobuf type information.

This could cause attacker-controlled strings to be emitted into generated JavaScript code.

Impact

An attacker who can first trigger a prototype pollution vulnerability may be able to influence generated `protobufjs` encode or decode functions in a way that can lead to arbitrary JavaScript execution.

This issue requires a separate prototype pollution primitive before `protobufjs` is invoked.

Applications without a reachable prototype pollution primitive are not directly exploitable through this issue alone.

Preconditions

- The application or one of its dependencies must allow an attacker to pollute `Object.prototype`.
- The polluted property must affect `protobufjs` internal type lookup behavior.
- The application must use `protobufjs` functionality that generates encode or decode code for affected types.
- The generated code path must be reached after the prototype pollution has occurred.

Workarounds

Avoid running affected versions in applications where attacker-controlled input can pollute `Object.prototype`. If immediate upgrade is not possible, remove or mitigate reachable prototype pollution primitives and isolate schema/message processing from untrusted application state.

References

- <<https://github.com/protobufjs/protobuf.js/security/advisories/GHSA-75px-5xx7-5xc7>>
- <<https://github.com/protobufjs/protobuf.js/releases/tag/protobufjs-v7.5.6>>
- <<https://github.com/protobufjs/protobuf.js/releases/tag/protobufjs-v8.0.2>>
- <<https://nvd.nist.gov/vuln/detail/CVE-2026-44291>>
- <<https://github.com/advisories/GHSA-75px-5xx7-5xc7>>

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